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MapOut

Week 3: Intelligent Agents and PEAS

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1. PEAS Specification for MapOut

MapOut is an AI-powered travel planner that aims to help users organize trips based on their interests, budget, and available time. Instead of manually searching for attractions and planning each activity, users will be able to enter their preferences and receive a personalized travel itinerary.

For the initial version, I decided to focus on Ohrid, Macedonia, because it offers different types of activities, including historical attractions, beaches, nature, and outdoor activities. This will allow me to test how the system responds to different user preferences before expanding to other cities.

MapOut can be considered an **intelligent agent** because it receives information about the user's requirements and its environment, evaluates possible activities, and makes decisions to achieve a specific goal. Its goal is to generate a suitable itinerary while considering the user's interests, available time, and budget.

1.1 PEAS Specification

To describe how the agent works, I used the PEAS model, which stands for **Performance Measure**, **Environment**, **Actuators**, and **Sensors**.

Component	Description
P – Performance Measure	Generate relevant recommendations, stay within the user's budget, create realistic schedules, minimize unnecessary travel between locations, and satisfy user preferences.
E – Environment	Tourist attractions and activities in Ohrid, their geographical locations, estimated prices, visiting durations, opening hours, and seasonal availability.
A – Actuators	Recommend suitable attractions, generate personalized itineraries, organize activities into daily schedules, and display estimated costs.
S – Sensors	User inputs such as destination, budget, interests, trip duration, and travel dates, together with information retrieved from the attraction dataset.

1.2 Explanation of the PEAS Components

Performance Measure (P)

The performance measure describes how successful MapOut is at generating an itinerary. A good itinerary should include activities that match the user's interests without exceeding their budget or available time.

For example, if a user has a budget of €50, the system should avoid recommending activities whose combined estimated costs exceed that amount. It should also consider the distance between attractions so the user does not waste too much time traveling.

Environment (E)

The environment includes the attractions and activities available in the selected destination. For the first version, this will be Ohrid. Some examples include Samuel's Fortress, the Ancient Theatre, the Church of St. John at Kaneo, beaches, and boat trips on Lake Ohrid.

Each attraction will have information such as its category, location, estimated cost, and visiting duration. This information will help the agent decide which attractions are suitable for the user.

Actuators (A)

Actuators represent the actions that MapOut can perform. Since MapOut is a software agent, its actions will be displayed through the application rather than performed physically.

For example, after evaluating the available attractions, MapOut can recommend several locations, organize them into a daily schedule, and display the estimated total cost.

Sensors (S)

Sensors represent how the agent receives information. In MapOut, the main sources of information will be the user's inputs and the attraction dataset.

For example, a user might enter Ohrid as the destination, select nature and adventure as interests, choose a two-day trip, and set a budget of €100. MapOut will use these inputs to decide which activities to recommend.

2. Initial System Architecture Diagram

The system architecture describes how the different components of MapOut will work together to generate personalized itineraries. For the initial version, I designed a simple architecture consisting of four main components: User Input, MapOut Intelligent Agent, Attraction Dataset, and Generated Itinerary.

2.1 Main Components

User Input

The user provides information about the trip, including the destination, budget, interests, trip duration, and travel dates. For example, a user might select Ohrid, choose a two-day trip, enter a €100 activity budget, and select history and nature as interests.

MapOut Intelligent Agent

This is the main component responsible for processing the user's preferences and selecting suitable activities. The agent will first identify attractions that match the user's interests. It will then evaluate constraints such as budget, available time, and location before organizing the selected activities into an itinerary.

In later weeks, I plan to implement search and planning algorithms to improve how the agent makes these decisions.

Attraction Dataset

The dataset will contain information about attractions and activities in Ohrid. Each attraction will have attributes such as its name, category, location, estimated price, visiting duration, and seasonal availability.

For example, Samuel's Fortress would belong to the history category, while kayaking would belong to the adventure category. The agent will use this information to compare activities and select suitable options.

Generated Itinerary

The final output will be a personalized travel itinerary showing recommended attractions, their planned order, estimated visiting durations, and total activity cost. For longer trips, the system will organize activities across multiple days.

2.2 System Architecture Diagram

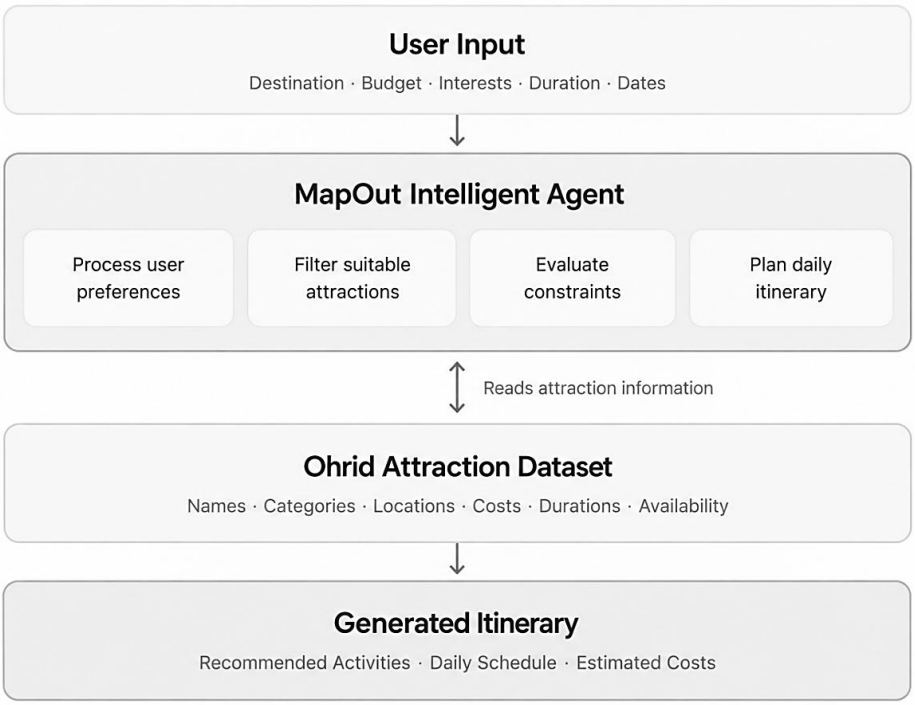


Figure 1. *Initial system architecture of MapOut, showing how user preferences and attraction data are processed to generate a personalized travel itinerary.*

2.3 Example of How the System Will Work

To better explain the architecture, I considered a traveler planning a two-day trip to Ohrid. The user enters a €100 activity budget and selects history and nature as their main interests.

First, MapOut receives these preferences. Then, it searches the attraction dataset for activities that match the selected categories. After that, it evaluates their estimated costs, visiting durations, and locations.

Based on this information, the agent will organize suitable attractions into a two-day itinerary. For example, the first day could focus on historical attractions in Ohrid's Old Town, while the second day could include nature-related activities around Lake Ohrid.

This is the planned workflow. The actual algorithms will be developed and tested in the following weeks.

3. Task-Environment Description

The task environment describes the conditions in which an intelligent agent operates and how these conditions influence its decisions. For MapOut, the environment consists of attractions, activities, locations, prices, and other information needed to plan a trip.

The real-world travel environment can change, but for the initial prototype, I plan to use a fixed dataset to make the system easier to implement and test.

3.1 Task-Environment Characteristics

Characteristic	Classification	Explanation
Observability	Partially observable	The agent may not have complete information about weather, attraction availability, or transportation conditions.
Number of agents	Single-agent	MapOut is the main decision-making agent responsible for generating the itinerary.
Determinism	Deterministic (initial prototype)	The same inputs and fixed dataset should produce the same results.
Episodic / Sequential	Sequential	Selecting one attraction affects the remaining time, budget, and possible next activities.
Static / Dynamic	Static (initial prototype)	The dataset will remain unchanged during itinerary generation, although real-world conditions may change.
Discrete / Continuous	Mixed	Attractions are discrete choices, while costs, travel distances, and visiting durations have numerical values.
Known / Unknown	Known (initial prototype)	The agent will use predefined attraction information and planning rules.

Although the real travel environment is partially observable, the initial prototype will use predefined attraction information. This allows the agent to work with the available data without requiring real-time updates.

3.2 Explanation with Examples

Partially Observable

MapOut may not have access to all information about the travel environment. For example, a boat trip might be unavailable because of bad weather, even if the dataset lists it as an available activity. Without real-time information, the agent cannot know this with certainty.

For the initial version, I will use predefined information and consider adding real-time data in later versions.

Single-Agent

MapOut will operate as a single decision-making agent. The user provides preferences, but MapOut is responsible for evaluating the available attractions and generating the itinerary. Other travelers or travel-planning agents will not participate in the decision-making process.

Deterministic

For the initial prototype, I plan to make the planning process deterministic. This means that if the user enters the same preferences and the dataset remains unchanged, MapOut should generate the same itinerary.

For example, two tests using Ohrid, a €50 budget, one day, and history as the selected interest should produce identical results. This will make it easier to compare different planning algorithms.

Sequential

Travel planning is a sequential problem because every selected activity affects the remaining options. For example, if a traveler has six hours available and spends two hours visiting Samuel's Fortress, only four hours remain for other activities and travel. The agent must consider previous decisions when selecting the next attraction.

Static / Dynamic

The real travel environment is dynamic because weather, prices, opening hours, and activity availability can change. However, the initial MapOut prototype will use a static dataset. The information will remain fixed while the agent generates an itinerary.

I chose this approach because it will allow me to focus on developing and testing the planning algorithm before adding more complex features.

Discrete / Continuous

MapOut's environment contains both discrete and numerical elements. Attractions are discrete because the agent chooses between specific activities, such as visiting Samuel's Fortress or taking a boat trip.

However, the agent must also consider numerical values such as ticket prices, travel distances, and visiting durations. For example, selecting an attraction that costs €15 will reduce the remaining activity budget by that amount.

Known / Unknown

The initial environment will be considered known because the agent will use predefined information about attractions and planning rules. For example, if the dataset specifies that an attraction takes approximately one hour to visit, MapOut will use that value when preparing the itinerary.

In reality, visiting durations and travel times may vary, but the initial prototype will rely on estimated values.

3.3 Why I Chose This Environment

I decided to start with a simplified environment because implementing a complete travel planner with real-time information would be too complex for the first version. By using a fixed dataset of Ohrid attractions, I can focus on understanding how intelligent agents make decisions and how search and planning algorithms can be applied to a real problem.

Once the basic system works, I can improve it by adding more destinations, more detailed activity information, and potentially real-time conditions.

CONCLUSION

During Week 3, I focused on defining MapOut as an intelligent agent and understanding how it will interact with its environment. I prepared the PEAS specification, designed the initial system architecture, and described the task-environment characteristics using examples from Ohrid.

For the initial version, I decided to focus on one destination and use a fixed attraction dataset to keep the project manageable. In Week 4, I plan to prepare the dataset, define the problem representation, and begin implementing the basic functionality in Python.

AI USE DISCLOSURE

I used ChatGPT (OpenAI) to help me understand the concepts of intelligent agents, PEAS, and task-environment characteristics. It also helped me review the system architecture, improve the wording and organization of my documentation, and create the architecture diagram.

I reviewed the explanations and suggestions, adapted them to my MapOut project, and made the final decisions about the project scope and design. I also made sure I understood the concepts and could explain how they apply to my project.